

Ahmed Mahmoud

Bachelor's degree

System and Biomedical Engineering

Faculty of Engineering, Cairo University

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GitHub Profile

LinkedIn Profile

Portfolio Website

Education

●Bachelor's degree in System and Biomedical Engineering

2023-2025

Faculty of Engineering - Cairo University

GPA: 3.3

Personal Projects

●Sociopedia - Social Media Platform

[github](#)

Dynamic full-stack social media platform

- Built using Node.js, Express.js, React.js, Material-UI, and MongoDB
- Implemented JWT-based authentication for secure user access
- Utilized Redux for efficient state management

●ChatEase - Real-time Chat Application

[github](#)

Full-featured messaging platform with real-time capabilities

- Developed using React, Node.js, Socket.io, and MongoDB
- Implemented JWT-based authentication, real-time messaging, and media uploads
- Created responsive UI with Tailwind CSS and seamless communication with Socket.io

●Learnova

[github](#)

Educational platform for interactive learning

- Developed a scalable web application to facilitate online learning
- Integrated features for course management, quizzes, and user progress tracking
- Utilized modern web technologies for a responsive and engaging user experience

●MYDB

[github](#)

Simple SQLite implementation

- Simple SQLite-like database implementation written in C. It demonstrates fundamental database concepts with an educational focus on how databases work internally.

●Sampling-Studio

[github](#)

Interactive tool exploring the Nyquist-Shannon sampling theorem

- Features real-time signal sampling, reconstruction, and frequency analysis
- Tools for signal mixing, noise control, and comparison of reconstruction methods
- Provides interactive environment for learning signal processing fundamentals

●TuneSpy

[github](#)

Python application for audio analysis and fingerprinting

- Generates spectrograms and extracts MFCC features from audio files
- Compares loaded audio with a preprocessed database to find similar matches

Additional Projects

●FilterLab - Digital Filter Designer

[github](#)

Powerful application for designing and visualizing digital filters

- Built using PyQt5 and Matplotlib for interactive filter design

- Supports standard filters and all-pass filters with visualization
- Generates corresponding C code for designed filters
- **BeamForming** *Python-based simulator with PyQt5 GUI for visualizing antenna array patterns* [github](#)
 - Real-time heatmaps and beam profiles with adjustable parameters
 - Interactive visualization of antennas, distance, delay, and frequency effects
- **SignalScope** *Desktop application for biomedical signal visualization and analysis* [github](#)
 - Multi-signal visualization with customizable controls
 - Signal processing tools with a user-friendly interface

Experience & Participations

- **Backend Mentor** *Jan - Apr 2025*
(EMBS) Student Branch Chapter at Cairo University *2025*
 - Mentored students in backend development technologies
 - Conducted workshops and hands-on sessions for Node.js and Express.js
 - Supported students in building their first backend applications
- **NASA SpaceApps Hackathon - Exosky Website** *Oct 2024*
Team Developer *Oct 2024*
 - Developed Exosky, a web application exploring exoplanets and stars using NASA APIs
 - Created interactive 3D visualizations, search functionality, and a drawing mode for constellations
 - Collaborated in a team environment to deliver a complete project under time constraints

Courses and Certificates

- **Alx_africa AI Career Essentials** (AICE) program in partnership : it focus on provide experience with practical AI tools and professional skills essential for career advancement
- **IBM Machine Learning with Python (Coursera)** : it involving on of Machine Learning using Python, honing my skills in data analysis, model evaluation, and predictive modeling techniques.
- **Supervised Machine Learning with DeepLearning.AI** (Coursera) : It provide key concepts in supervised learning, particularly in building predictive models using linear and logistic regression.

Technical Skills

Backend: Node.js, Express.js, MongoDB, PostgreSQL, Jest

Frontend: React.js, Tailwind CSS, Redux, Material UI

Desktop Application: Python, PySide6, PyQt5, Matplotlib

General: C++, Pandas, NumPy, OOP Programming, scikit-learn

Other Tools: Git, GitHub, Socket.io

Areas of Interest: Web Development, Signal Processing, Desktop Applications

Languages

- **Arabic:** Native/Fluent
- **English:** Intermediate (Reading and Writing)